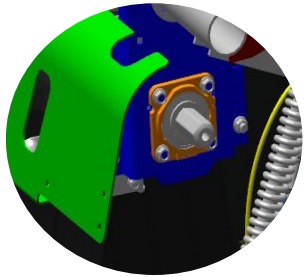
MACHINE HEALTH MONITORING SYSTEM FOR BALER BEARINGS

University of Minnesota – Twin Cities x
John Deere

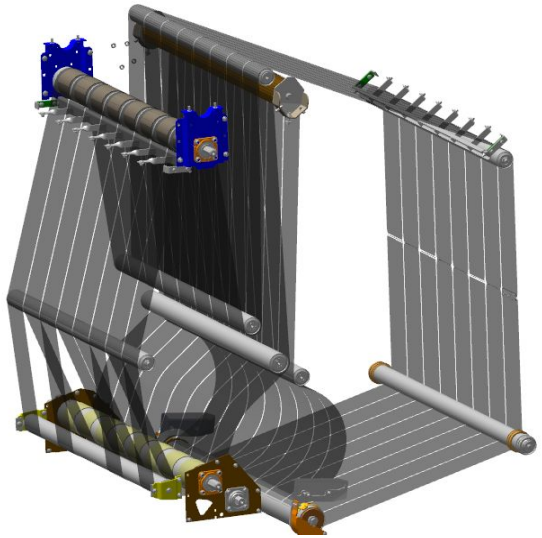


PROJECT IDENTIFICATION

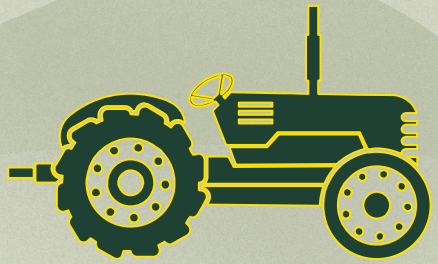


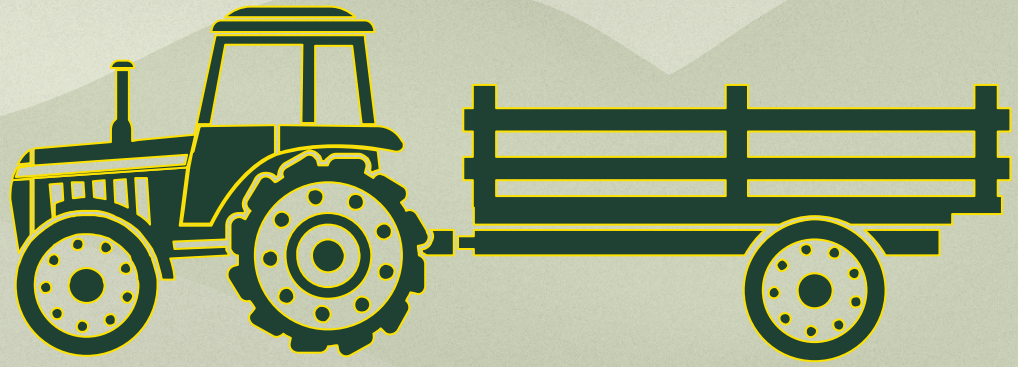
- Partnered with John Deere to design a health monitoring system to ideally be implemented with baler bearing production

- Objective: decrease baler component failures by making users aware of potential hazards using bearing health data



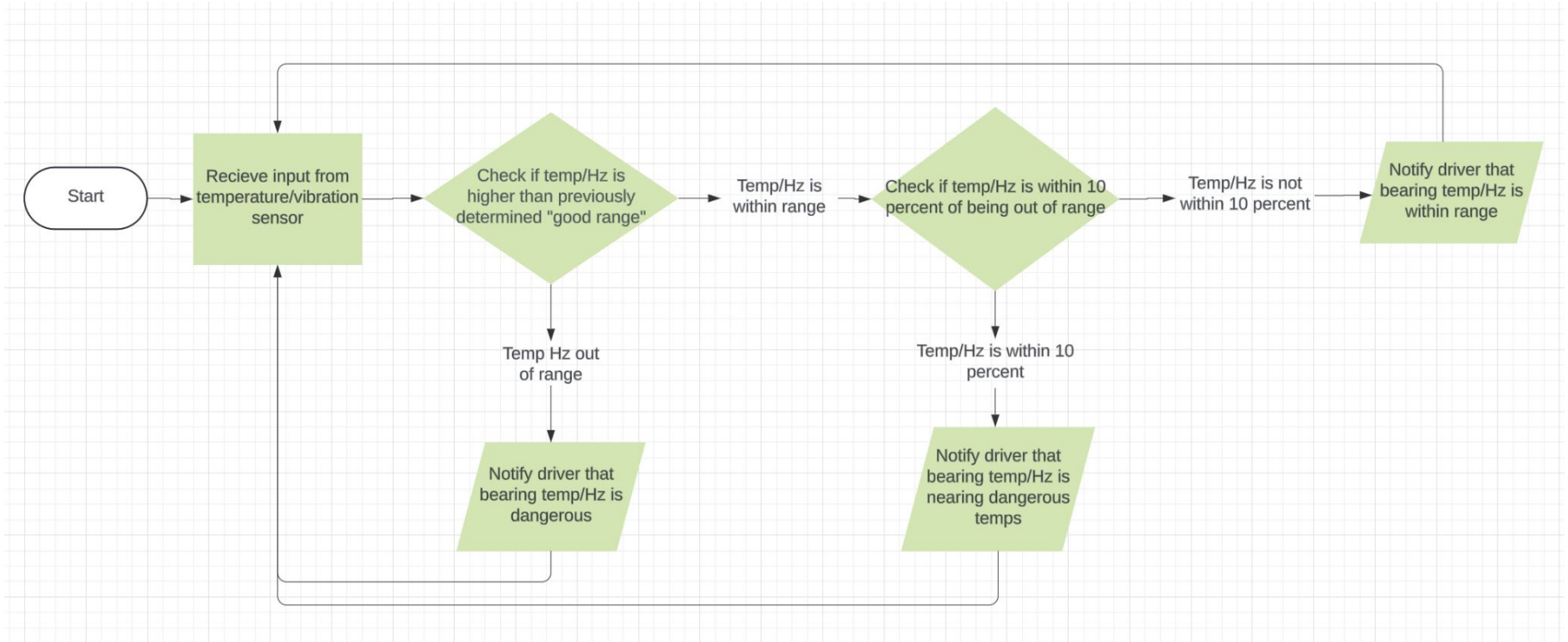
01 PROJECT GOAL





OVERVIEW

- Without proper care and attention, bearings can heat up from excessive use and potentially set a fire to the dry hay
- Problem: It is difficult for the baler operator to keep track of the bearing's health while it is in operation
- Solution: develop a mechanism that evaluates the temperature of the bearing continuously and updates the operator in real time about potential hazards


Start
Recieve input from temperature/vibration sensor
Check if temp/Hz is higher than previously determined "good range"
Temp/Hz is within range
Check if temp/Hz is within 10 percent of being out of range
Temp/Hz is not within 10 percent
Notify driver that bearing temp/Hz is within range
Temp Hz out of range
Notify driver that bearing temp/Hz is dangerous
Temp/Hz is within 10 percent
Notify driver that bearing temp/Hz is nearing dangerous temps

BUSINESS CASE: DEFINING THE PURPOSE

What benefits to John Deere **and** their customers can our ideas produce?

- Improved longevity of baler bearings due to early detection of failures, resulting in lower maintenance costs for customers
- Higher overall product efficiency for balers, giving Deere the competitive edge on the market
- Increased product marketability due to lower insurance costs since bearings can now be more easily maintained

COLLABORATING WITH JOHN DEERE

- **Biweekly** video calls over Zoom to share project updates and receive feedback
- Supplied all project materials: hexagonal bore ball bearings, electronic components, display tablet
- Staying in touch with Deere advisors to schedule delivery of prototype
- Went on a site visit to John Deere's factory to learn more about the baler processes and to hear from the advisors and their teams





REQUIREMENTS SUMMARY

Functional requirements and constraints

- ❏ The monitoring system notifies the baler operator when a bearing is nearing unhealthy conditions in a prompt and clear manner
- ❏ Accommodate any outside conditions in which the baler is operating at
- ❏ Prototype system can be seamlessly integrated into current baler systems
- ❏ Technologies and sensors employed in the prototype must be cost-effective to mass produce

02 TEAM MEMBERS

MEET THE TEAM

KAVYA PRASAD



JANANI KANNAN



ANYA BHOWMICK

JULIA GROSS

SANAB ABDI

SOPHIE REZNIK



SMRITI SAMTANI



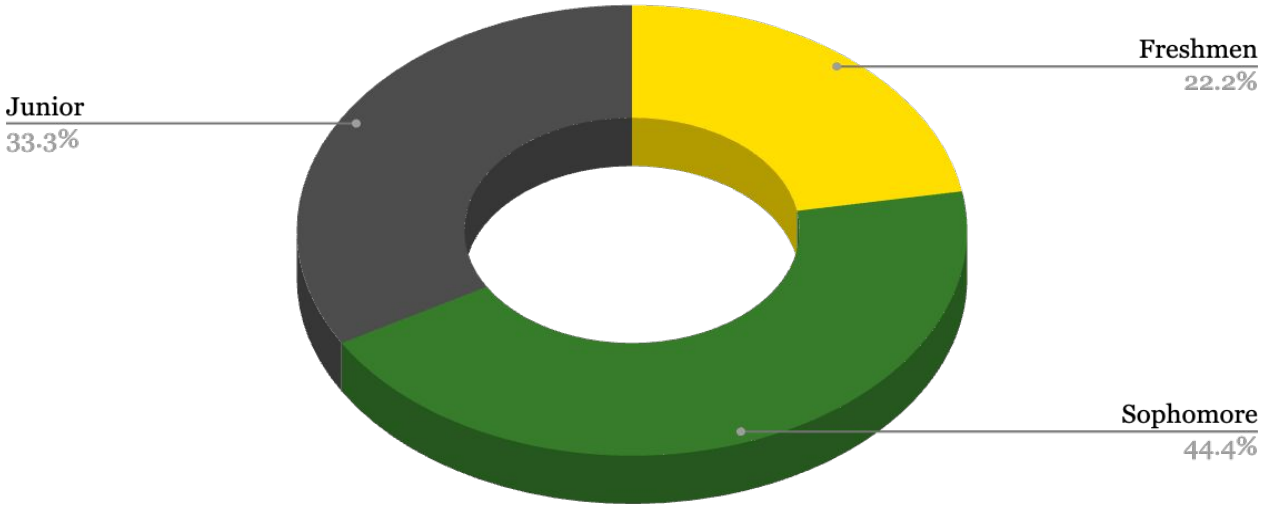
AISHWARYA RAMKUMAR



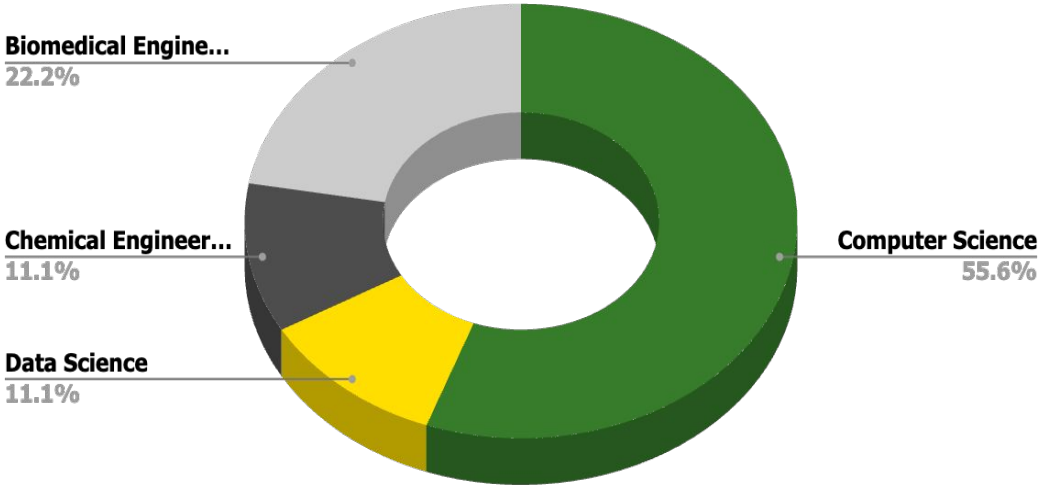
AKANKSHA RAI

03 TEAM BREAKDOWN

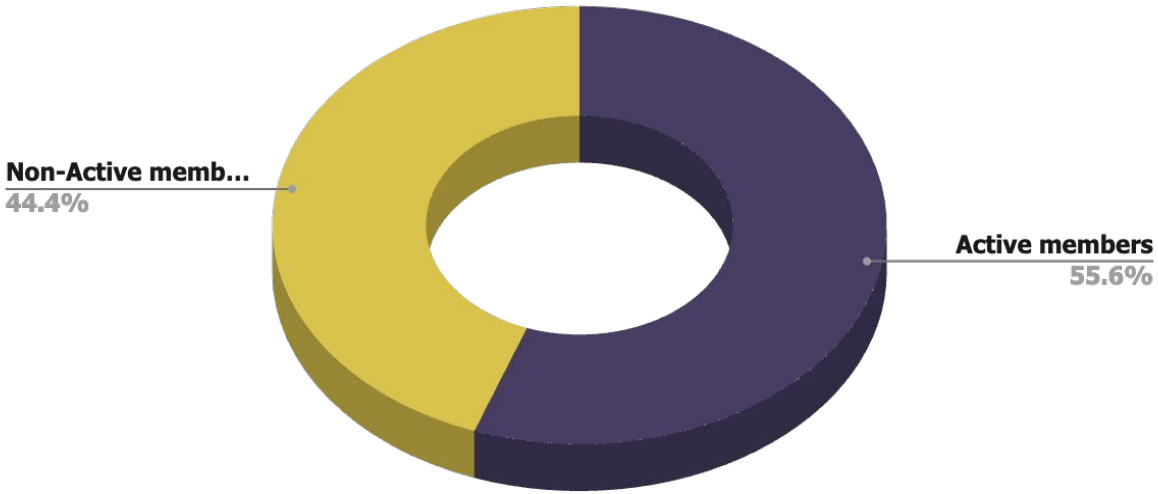
Year in School



Majors of the Group



SWE Membership Status



04 TEAM WORK

TEAM WORK

WEEKLY TEAM MEETINGS

- Focus-group meetings (Product Design and Software Design)
- Full-team meetings
- Multiple workdays during the build phase

TEAM BUILDING ACTIVITIES

- Site visit (John Deere, Ottumwa, Iowa)
- Ice Breakers, Hands-on Workshop sessions

COMMUNICATION

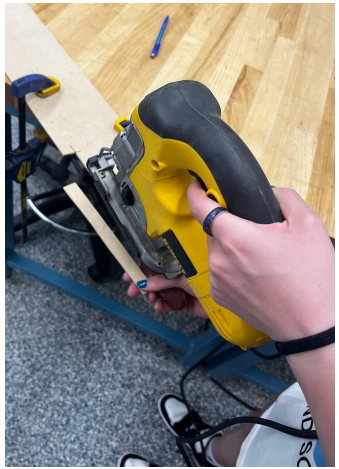
- Slack group chat, Emails, Text message

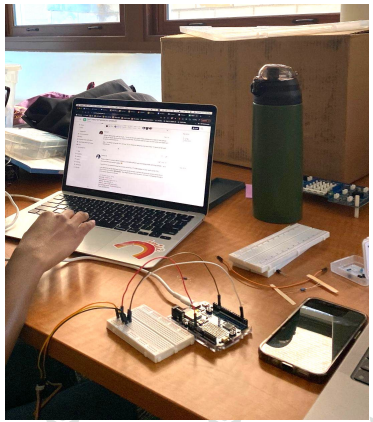
GOAL SETTING

- SMART Goals

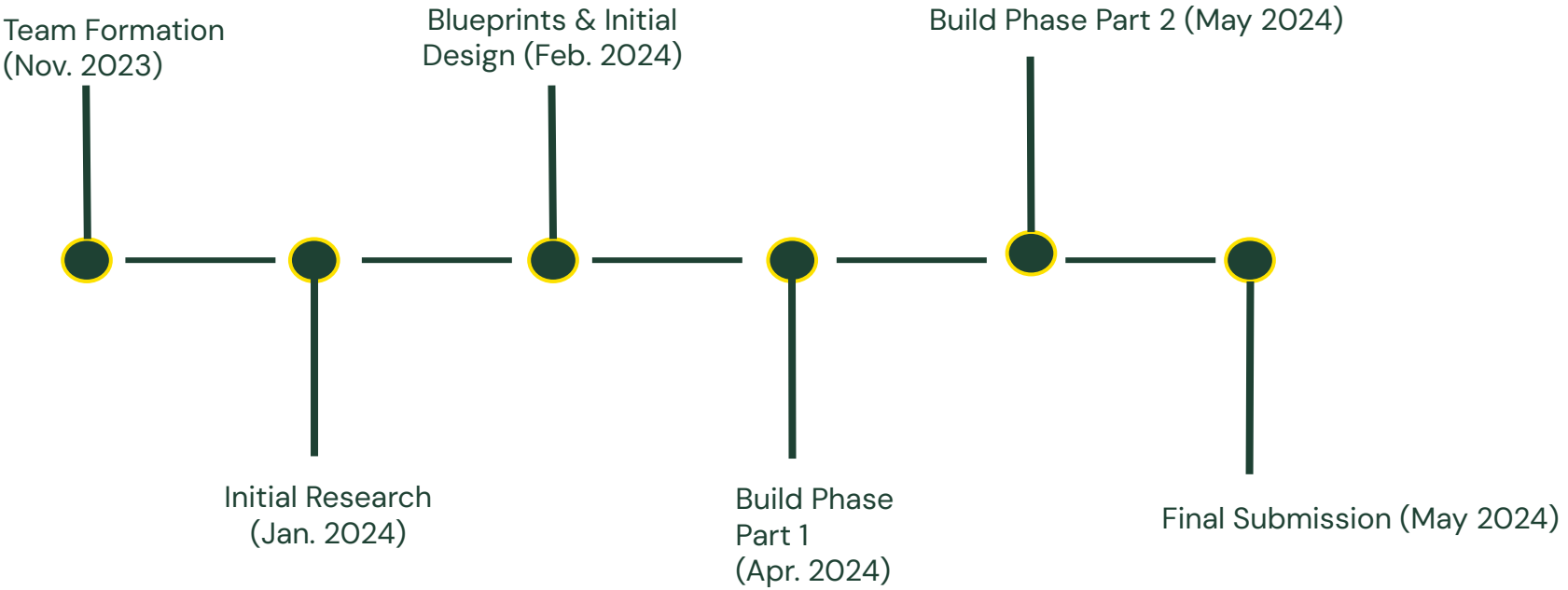








05 TIMELINE


Team Formation
(Nov. 2023)
Initial Research
(Jan. 2024)
Blueprints & Initial
Design (Feb. 2024)
Build Phase
Part 1
(Apr. 2024)
Build Phase Part 2 (May 2024)
Final Submission (May 2024)

06 BUDGET

BUDGET

Item	Cost
ROUND 1 OF ORDERS	
Arudino Sensor Kit	$62.30
Bearing	$58.86
Arduino UNO R4 Enclosure	$19
Arduino UNO R4	$28.50
Wires	$6.98
Breadboard	$9.79
Arduino Piezo Vibration Sensor Module	$6.99
Starter kit for Arduino	$11.99
Temperature probe-like sensors	$9.99
Arudino temperature sensors	$9.99
ROUND 2 OF ORDERS	
Bearing Dust Shield for Balers	$8.99
Screws	$6.99
Screwdriver Set	$9.99
Sensor bearing mount	$14.98
ROUND 3 OF ORDERS	
Tablet/display for demonstration	$239.99
Terminal Adapter for Temperature Sensors	$16.99
Tape	$2.78
USB Type C Cable	$6.99
Grove 3-Pin Connectors	$7.29
Temperature probe-like sensors	$9.99
Wires	$6.98
TOTAL	$556.35

- Effectively utilized only 15% of the allocated budget for the project, making the prototype cost-effective to produce in mass.
- Made orders in rounds, to avoid duplication of resources and make more informed decisions about the essential resources we will need.

SITE VISIT COSTS	
Expense type	**Cost**
Hotel	$725.76
Meals	$356.71
Mileage	$451.58
TOTAL	$1,534.05

07 INITIAL RESEARCH PHASE

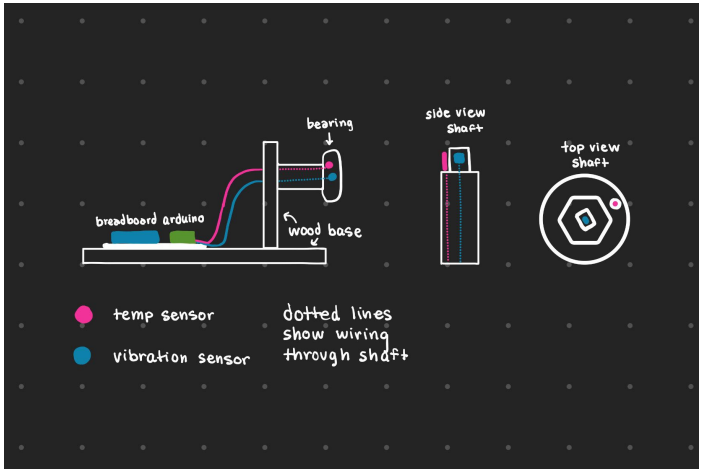
BALER RESEARCH

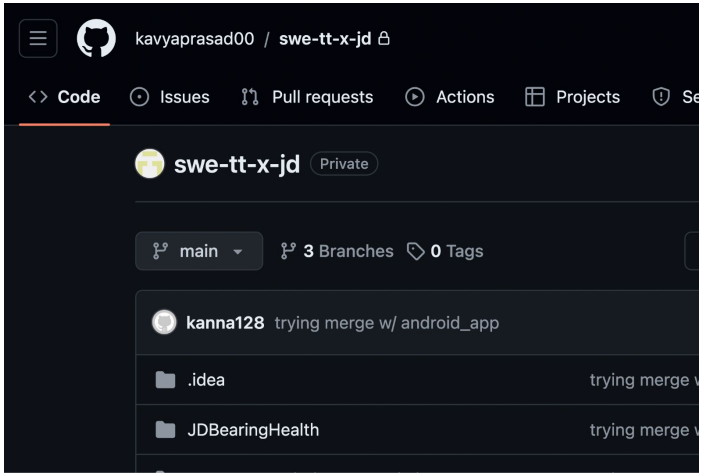
- Baler belts
  - Belts rotate to compact the hay
- Baler chains
  - Low tension can wear down sprockets faster
  - Oiling chains can increase their lifetime
- Net wraps
  - Wraps plastic around finished bales for storage
- Pick-up teeth
  - Picks up hay and smooths it out
  - Damaged or crooked teeth can decrease efficiency
- **Bearings**: **Chosen Subcomponent**
  - Ensures smooth movement of shafts
  - Cracked bearings are a massive structural integrity failure and is dangerous to the farmer
  - Bearings overheated from friction can cause fires
  - Bearings are difficult to monitor, especially while the baler is in active use



08 BUILD PHASE

FOCUS GROUPS





Product Design

Software Design

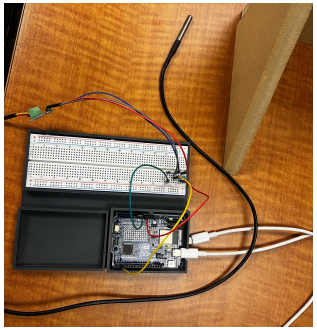
PART 1: 4 SPRINTS

Product Design Focus Group

- Shaft design blueprint evaluations
- Finalize part assembly
- Multiple iterations for the 3D printed shaft
- Initial brainstorming of the custom base
- Creating wood support blueprints

Software Design Focus Group

- Collect and analyze data from sensors
- Integrate hardware and software components
- Unit test Arduino IDE code
- Debugging electronic components





PART 2: 2 SPRINTS

Product Design Focus Group

- Building wood support and custom Arduino + breadboard base
- Adding and removing parts as necessary for design optimization

Software Design Focus Group

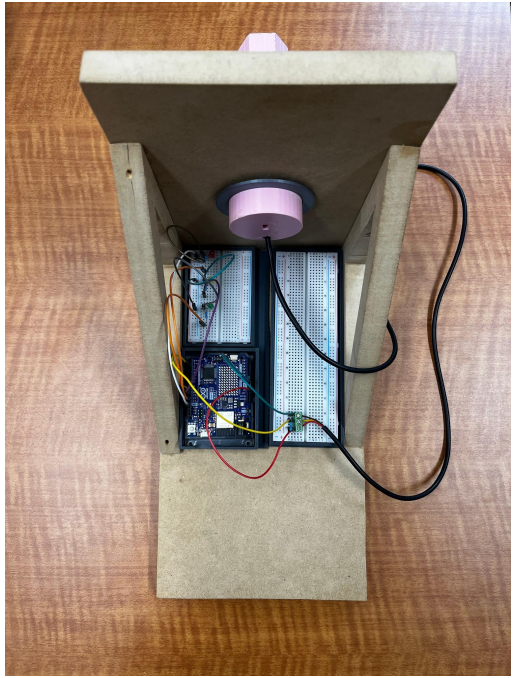
- Completing temperature sensor code
- Implement User Interface to display messages to the operator
- Integrate Arduino IDE program, python script, and Android Studio program

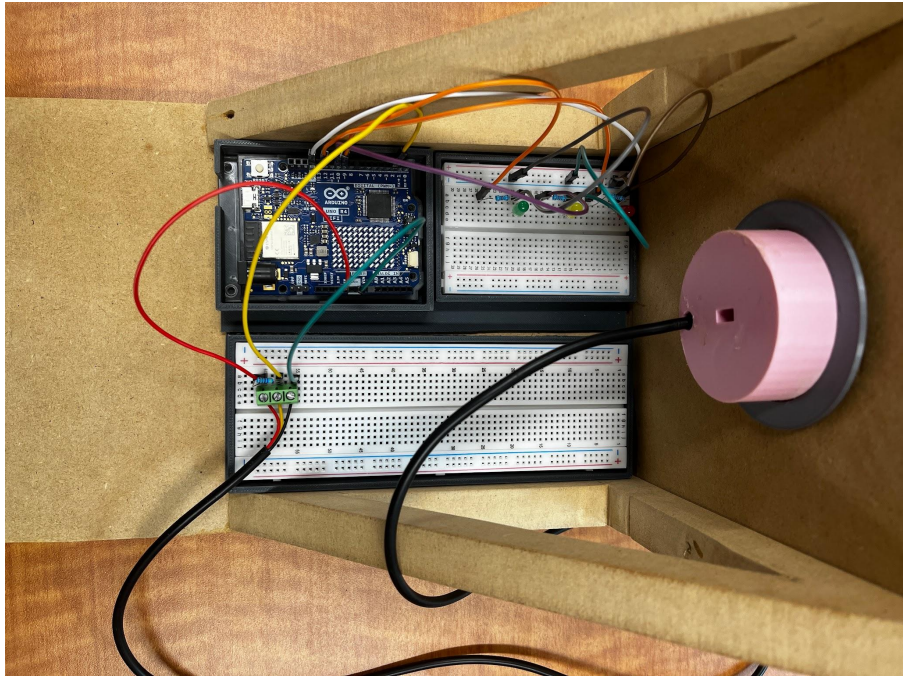
Both Groups

- Consult advisors for feedback
- System testing to ensure all components produce expected results in an integrated environment

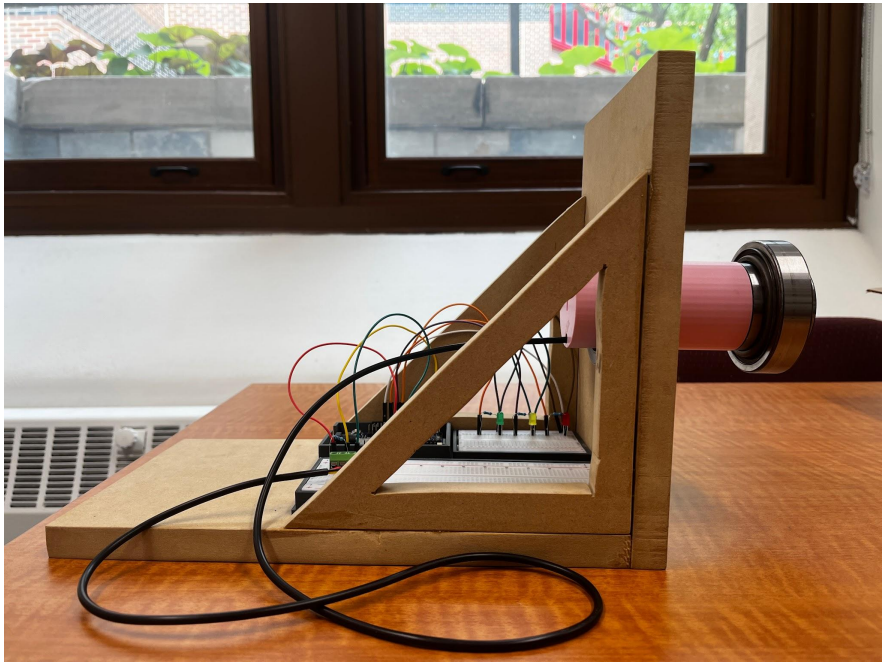
09 OUTCOMES

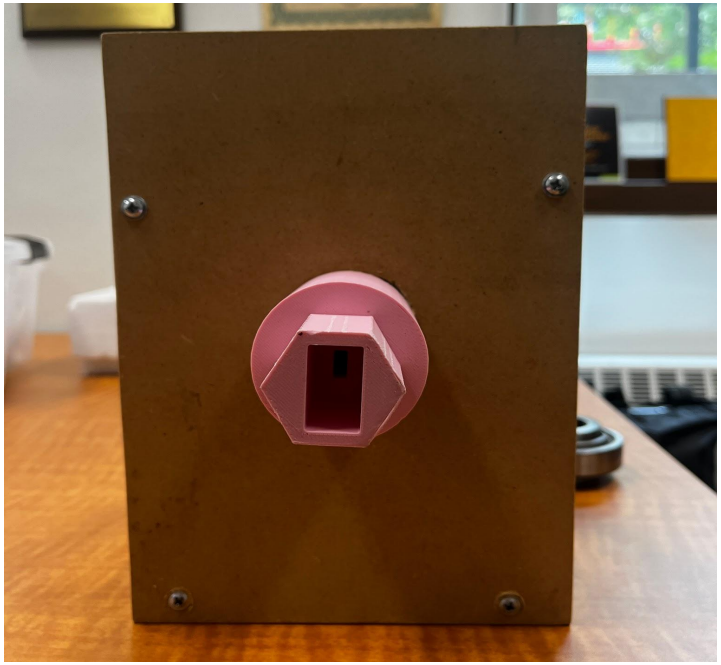
PROTOTYPE





PROTOTYPE

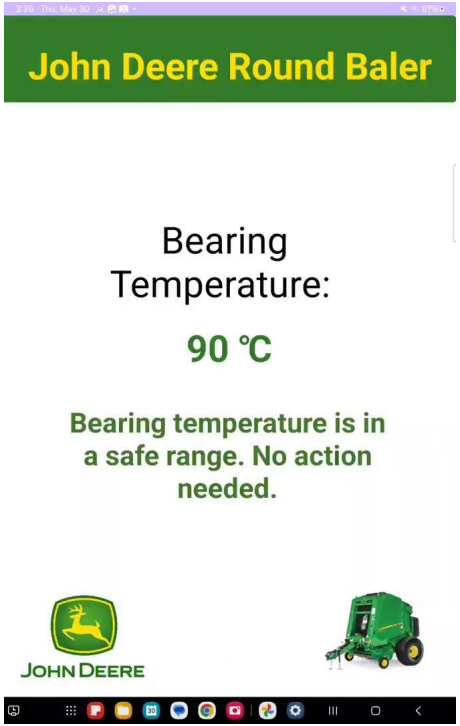




RESULTS SUMMARY

This is a demonstration of how our final project works. This is the application we developed displaying temperature values of the bearing, and corresponding output to alert the operator based on what range the temperature values fall under.

While the temperature sensor does send data to Firebase and generally is displayed on this application, the temperature values shown in this demonstration are test values read from Firebase for us to be able to showcase the output for all three cases of temperature ranges.



10 OBSTACLES OVERCOME

OVERCOMING OBSTACLES

Obstacles

1. Learning curve to learn new software and hardware concepts, such as Solidworks, Firebase, Arduino, and understanding electricals and wiring as well as the workings of the sensors from scratch
2. Getting exact measurements for the shaft design to precisely fit the bearing, sensors, and wood support was difficult
3. Trouble with getting the temperature sensor to read values and not overheat
4. Compatibility issues with the Arduino connecting to Firebase over WiFi, and issues with code libraries not working well with the Arduino

Overcoming these obstacles

1. Did extensive research during the design phase to learn about how the materials we ordered worked. Used videos and online documentation on the software to understand them. Ensured all team members found resources to learn
2. Did multiple iterations of the 3D printed shaft while modifying the design and measurements
3. Debugged the sensor code and swapped out the hardware (i.e. resistor) that work well with the sensor after doing research on the issue
4. Debugged code to find issue. Wrote multiple versions of the code and created a final Python script to manage the data to Firebase instead of using incompatible WiFi libraries. Changed software to work with hardware we had

QUESTIONS?